DEEP LEARNING COURSEWORK

Age estimation and Gender Classification using Convolutional Neural Networks

age gender A.keras
https://drive.google.com/file/d/1It5BApVPINWz6tUezKYE70wU4GKfMlxP/view?usp=sharing

age gender B.keras
https://drive.google.com/file/d/1O7gJr7d-FlQLfssrOkLv0xYkRYCKa09g/view?usp=sharing

University of Bath

2025-2026

Contents

Section 1: Introduction

The purpose of the coursework is to create two convolutional neural network-based models to classify the gender of the person and estimate the age of the person from the images provided from a small subset of the UTKFace dataset (UTKFace, n.d).

The UTKFace dataset contains more than 20000 images, but for our coursework, a smaller subset of 5000 images is being used, out of which two features are taken into consideration, that is, age and gender. The project's coursework involves creating two models

- Custom Model designed from scratch
- Pre-Trained CNN model where the transfer learning process has been used

Both the models are trained using Colab Pro(Carneiro et al., 2018), and the final models are saved as _.keras file, and their shareable links are mentioned above in the cover page

Section 2: Custom CNN Model

A. Data Augmentation Process

The Primary reason data augmentation was done to introduce real world problems like small changes in facial expressions, orientation of the images using techniques like random zoom and random rotation by small amount 5%, while doing little bit of testing like with other augmentation techniques like random saturation it would distort the images and that would affect the ability of the model to learn properly as well as applying minimum augmentation on a smaller dataset would lead to perfect results and the augmentation process was done dynamically rather than expanding the dataset, just shuffled, batched and prefetching the images for better training purpose, and overall use of augmentation was to generalise the model in a better was as a consequence of it improving the performance of model in predicting the age and classifying gender successfully (Team, n.d.).

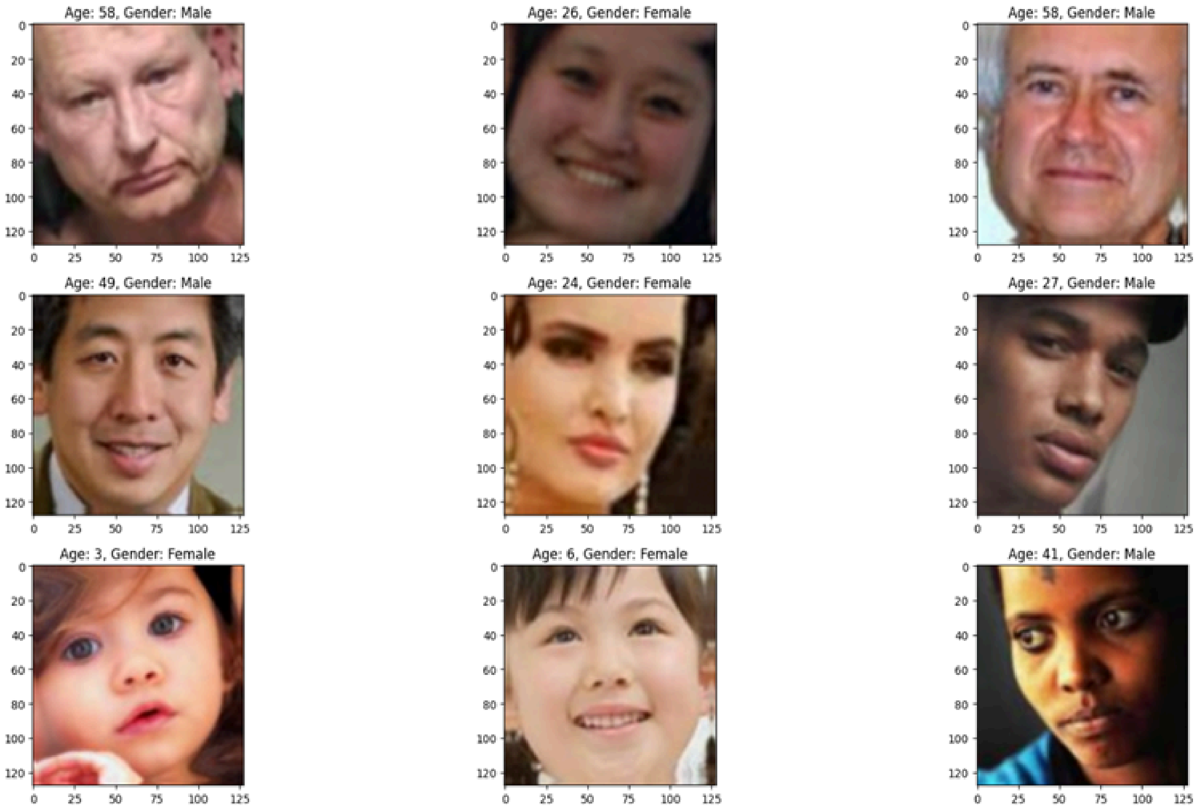

Figure 1: Data Augmentation Processed Images

B. Custom CNN Model Architecture

The custom CNN model created is being designed to serve the purpose of multi-task learning, enabling to prediction of age (regression) and classification of gender (binary classification) tasks. The Model is initialised with an input layer with an image of size (128*128*3) which is associated with an RGB images used for this task, then added 4 convolutional blocks to extract image features of increased filter size that is 32,64,128,256, Each Block had a Conv2D layers followed by a layer of batch normalisation to keep learning stable enough and for dimensionality reduction MaxPooling2D was used and dropout was used to solve overfitting (Srivastava et al., 2014).

The Model has a dense layer of 512 neurons that is a shared layer with a dropout as well to create a commonality for both tasks, then it had two separate branches with 256 neurons to predict a linear regression task age which used ReLu activation function, every time a negative integer appears, ReLU logic substitutes a 0 this keeps CNN model sound, A ReLU layer outputs data that has a size equal to the input's and is empty(Kumar et al., 2024), The gender classification follows the same route with sight changes like with 128 neurons and a sigmoid activation function better suited for classification)

The Model has a kernel initializer called "He Normal", which is used to prevent exploding gradients, and it is specifically designed for ReLU as a part of fine-tuning (TensorFlow, 2025), where we could see some spikes as a part of training

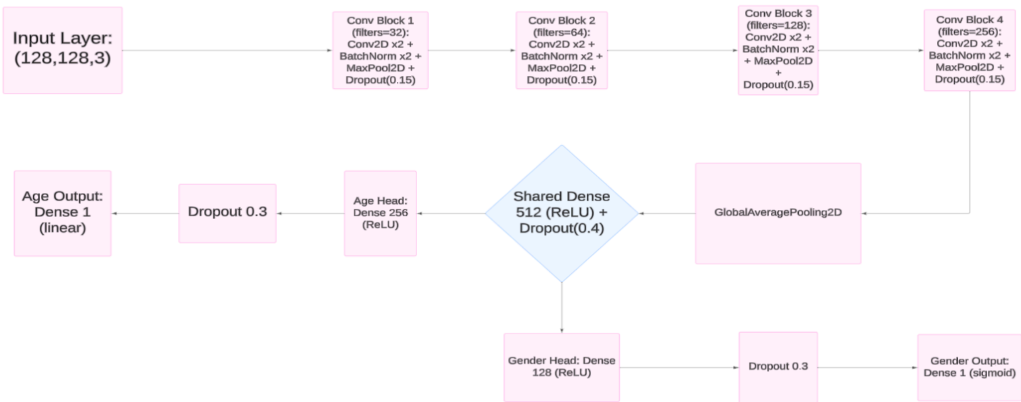

Figure 2: CNN Architecture Workflow

Here's the architecture plot in a shareable link:
MODEL A ARCHITECTURE.png

C. Training the data, Hyperparameters Usage, and Model Evaluation

Custom Model is trained with the Adam Optimizer with a learning rate of 1e-4. Adam is the recently proposed first-order gradient-based optimization method for stochastic objective functions (Xiong et al., 2019). , which is the same for both models used for this coursework. Both model uses two different loss functions to calculate loss for each parameter: MAE for age loss and Binary Cross-Entropy for Gender loss, to balance both tasks, loss weight parameters have been used as well

This model achieved a validation accuracy of 82.6% and training accuracy of 79.47% and the Age MAE for the validation dataset is 7.5, and for the training dataset is 6.23

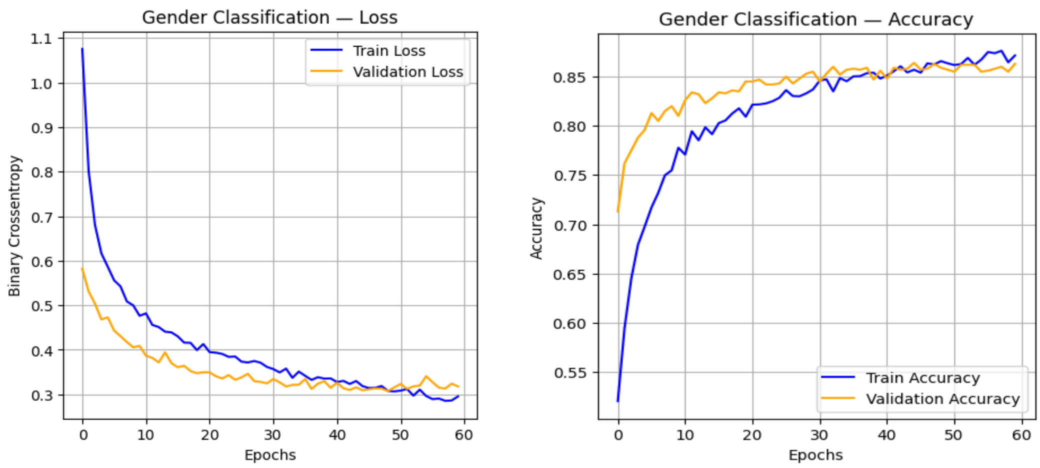

Figure 3: Gender Classification

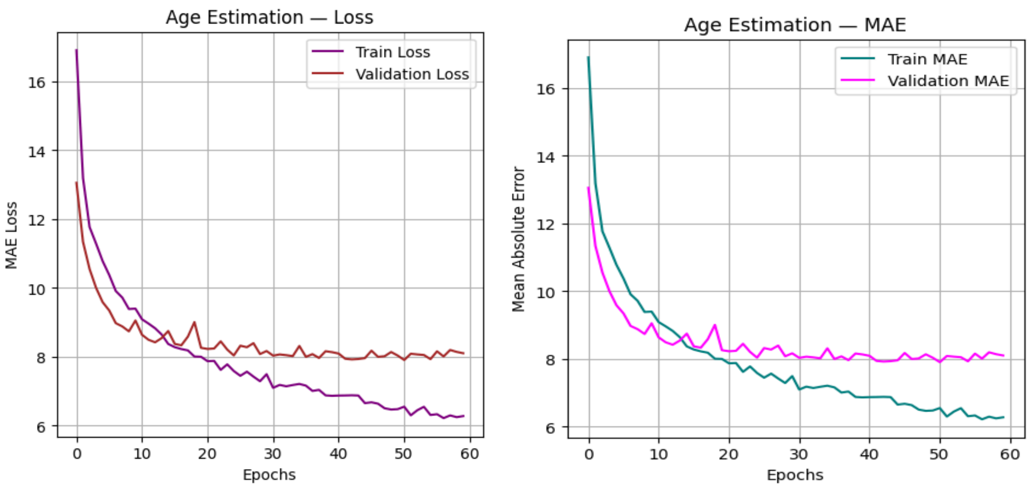

Figure 4: Age Estimation

Section 3: Pre-Trained CNN Model

A. Purpose of using the Pretrained Model

The purpose of using a pre-trained model like VGG16 is to use its feature extraction capabilities, as it's been trained on a larger dataset like ImageNet. By using this model, it could benefit from the pretrained features by freezing the weights of the layers and all, and it significantly reduced the learning time and improved the accuracy with smoother curves in the graph without any distortion(Kumar et al., 2024; Exxactcorp.com, 2025).

B. Model Architecture

The Pretrained CNN Model uses VGG16 as the base model for feature extraction. This model is loaded with ImageNet weights, and it excludes the classification layers using the include_top = False parameter. The model uses the same input images and processes them through the VGG16 model for feature extraction. After that, the architecture has a Flatten layer to convert the 2D vectors into a 1D vector, then a shared dense layer of 256 units and ReLu activation is added on top of that feature extractor to learn the representation, which can be useful for gender classification tasks and age estimation tasks simultaneously (Liu et al., 2024), and similar to Model A, dropout has been added. Then the model follows a similar path to Model A, with shared dense layers for age_branch, it is with 128 neurons with ReLu activation, and for gender_branch with the same number of neurons with Sigmoid activation function, and early stopping callbacks were initialised to get the best value

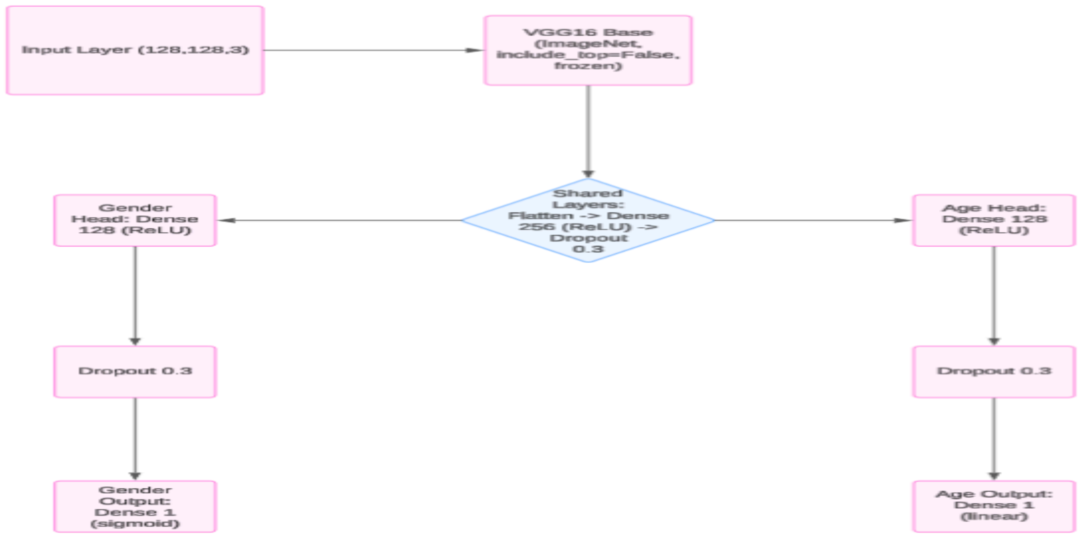

Figure 5: PreTrained CNN Architecture Workflow

Here's the architecture plot in a shareable link:

MODEL B ARCHITECTURE.png

C. Training the data and Hyperparameters Usage, and Model Evaluation

All the parameters involved in training for Model A were used in the pretrained Model as well, using the same Adam Optimizer and with the same learning rates as well, with slight changes in the hyperparameters, from loss_weights={"age_output": 0.7, "gender_output": 0.3} to loss_weights={"age_output": 0.6, "gender_output": 0.4}

This model achieved a validation accuracy of 86.95% and training accuracy of 88.19% and the Age MAE for the validation dataset is 7.9, and for the training set was 6.23

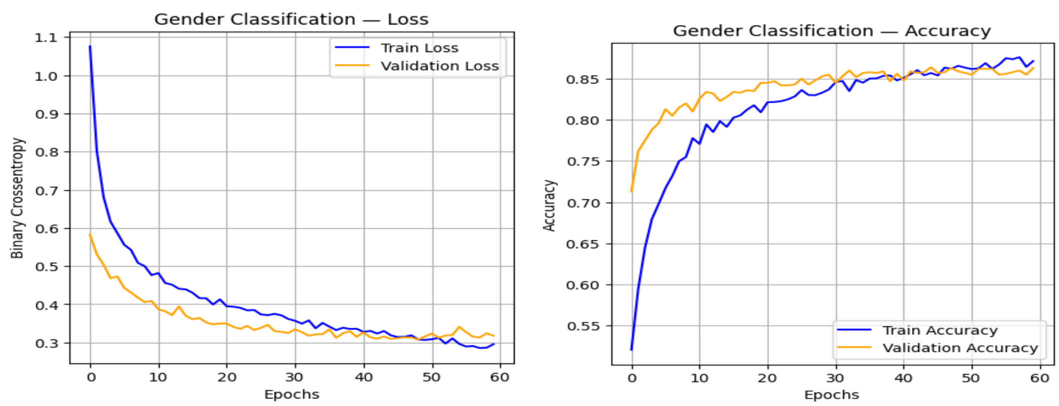

Figure 6: Gender Classification

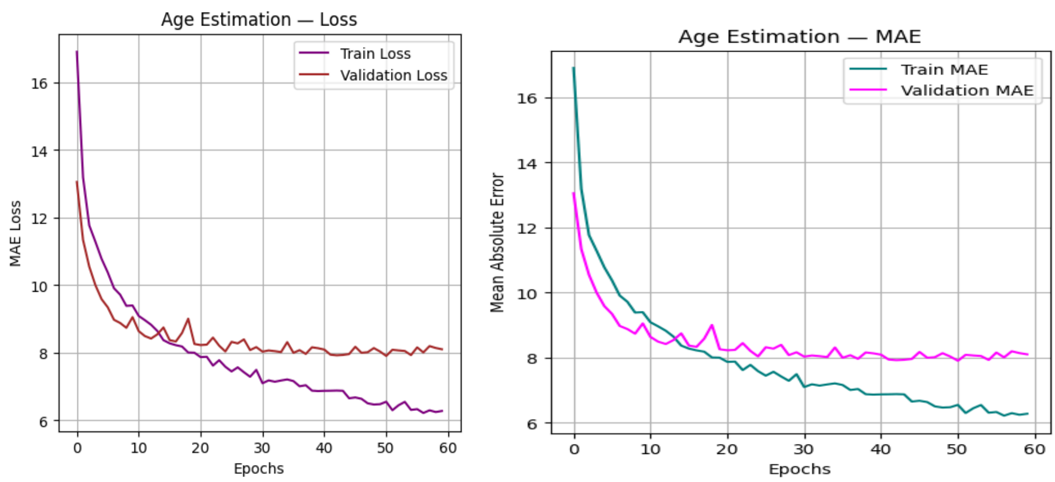

Figure 7: Age Estimation

Section 4: Summary and Discussion of the Models

A. Comparative Analysis

A custom CNN Model is built from scratch to learn features directly from the images through layers, and it also offers flexibility and complete control over the architecture. It requires longer training, results in decent performance with 82.61% gender classification accuracy, and 7.5 years MAE for age prediction. The only limitation comes from a smaller architecture, which was solved using a regularizer when compared to the modelB , VGG16 is pretrained on ImageNet for feature extraction, significantly boosting performance by the transfer learning process. This did reduce the training time as well, and it achieved validation accuracy of 86.95% and Age MAE of 7.9. The difference in AGE MAE is 1.6. Model B required more computational resources; the Pretrained CNN Model was good in terms of handling convoluted visual patterns and generalising better.

Table 1: Comparative Analysis of Model A and Model B

MODEL	MAE (Training)	MAE (Validation)	Accuracy (Training)	Accuracy (Validation)
MODEL A	5.9	7.5	79.47%	82.60%
MODEL B (VGG 16)	6.23	7.9	88.19%	86.95%

B. Pros and Cons of Custom CNN

Model A had a certain level of influence, as it shows its capacity despite being smaller as compared to Model B, as it was built from scratch, and it did help us to understand the concepts in a better way the only downside though it struggled initially to learn the model then with the help of regularizer it was able to learn which in contrast with pretrained model which could do it independently

C. Pros and Cons of PreTrained CNN

Model B showed the pros of transfer learning, particularly in gender classification, but there was one slight disadvantage due to the increase in the number of parameters it prone to overfitting. The comparison of these two models highlights a key information transfer learning is powerful; it's only effective and depends heavily on dataset size and the similarity factor between ImageNet features and the target domain. In contrast, a carefully designed custom network can match or outperform pretrained models under the constraints of small, specialised datasets.

D. Conclusion

In conclusion, this assignment provided an understanding of both custom CNN design and the transfer learning ecosystem. The experiments revealed the situations between model capacity, dataset size, training stability, and generalisation. The hands-on process of building, training, and evaluating two different architectures elaborated how deep learning models must be adapted thoughtfully to the characteristics of the dataset and the prediction tasks involved.

References

1. Carneiro, T., Medeiros Da Nobrega, R.V., Nepomuceno, T., Bian, G.-B., De Albuquerque, V.H.C. and Filho, P.P.R., 2018. Performance Analysis of Google Colaboratory as a Tool for Accelerating Deep Learning Applications [Online]. *IEEE Access*, 6, pp.61677–61685. Available from: https://doi.org/10.1109/access.2018.2874767.
2. Exxactcorp.com, 2025. *Freezing Layers in Deep Learning and Transfer Learning | Exxact Blog* [Online]. Available from: https://www.exxactcorp.com/blog/deep-learning/guide-to-freezing-layers-in-ai-models.
3. Kumar, R., Singh, K., Mahato, D.P. and Gupta, U., 2024. Face-based age and gender classification using deep learning model [Online]. *Procedia Computer Science*, 235, pp.2985–2995. Available from: https://doi.org/10.1016/j.procs.2024.04.282 [Accessed 21 October 2024].
4. Liu, S., Yue, W., Guo, Z. and Wang, L., 2024. Multi-branch CNN and grouping cascade attention for medical image classification [Online]. *Scientific Reports*, 14(1). Available from: https://doi.org/10.1038/s41598-024-64982-w [Accessed 5 January 2025].
5. Srivastava, N., Hinton, G., Krizhevsky, A., Ilya Sutskever and Ruslan Salakhutdinov, 2014. Dropout: A Simple Way to Prevent Neural Networks from Overfitting [Online]. *Journal of Machine Learning Research*, 15, pp.1929–1958. Available from: http://jmlr.org/papers/v15/srivastava14a.html.
6. Team, K., n.d. *Keras documentation: Image classification from scratch* [Online]. *keras.io*. Available from: https://keras.io/examples/vision/image_classification_from_scratch/.
7. TensorFlow, 2025. *tf.keras.initializers.HeNormal | TensorFlow v2.16.1* [Online]. Available from: https://www.tensorflow.org/api_docs/python/tf/keras/initializers/HeNormal?_gl=1 [Accessed 26 November 2025].
8. UTKFace, n.d. *UTKFace* [Online]. Available from: https://susanqq.github.io/UTKFace/.

9. Xiong, F., Xiao, Y., Cao, Z., Gong, K., Fang, Z. and Zhou, J.T., 2019. *Good practices on building effective CNN baseline model for person re-identification* [Online]. Available from: https://doi.org/10.1117/12.2524386 [Accessed 31 October 2024].